

StratAI - Lovable.dev Frontend Development Prompt

Project Overview

Build a modern, responsive web application for **StratAI** - a multi-agent strategic intelligence platform that helps hardware executives make data-driven decisions about semiconductor supply chain, GPU allocation, and competitive positioning.

Core Functionality

User Flow

1. **Landing page** with value proposition and CTA
2. **Strategic question form** to submit analysis requests
3. **Real-time analysis display** showing progress
4. **Report viewing** with confidence scores and recommendations
5. **AI chat interface** for follow-up questions
6. **Feedback submission** to rate reports
7. **Admin dashboard** for analytics (optional MVP feature)

Key Features

- Strategic analysis request form with market signal sliders
 - PDF upload for context enrichment
 - Real-time progress indicators during analysis
 - Interactive report display with sections
 - AI chat assistant for Q&A about reports
 - Feedback collection system
 - Responsive design (mobile, tablet, desktop)
-

Design Requirements

Brand Identity

Name: StratAI
Tagline: "Strategic Intelligence Powered by Multi-Agent AI"
Industry: B2B SaaS for Hardware/Semiconductor Executives

Color Palette:

- Primary: Deep blue/purple gradient (#667eea → #764ba2)
- Secondary: Professional navy (#2c3e50)
- Accent: Vibrant blue (#3b82f6)
- Success: Green (#10b981)
- Warning: Amber (#f59e0b)
- Error: Red (#ef4444)
- Neutral: Gray scale (#f8fafc to #1e293b)

Typography:

- Headings: Inter or Poppins (bold, modern)
- Body: Inter or System UI (clean, readable)
- Code/Data: JetBrains Mono or Fira Code

Design Style:

- Modern, professional, executive-focused
- Data-rich but not overwhelming
- Confidence through clean design
- Glassmorphism effects for cards
- Subtle animations for engagement
- High contrast for accessibility

Pages & Components

1. Landing Page (/)

Hero Section:



with explainable confidence scoring.
[Start Free Analysis →] [Watch Demo]
Trusted by executives at Azure, Google Cloud, Oracle, NVIDIA, Tesla

...

****Features Section:****

...

			
5 AI Agents	Confidence Scoring	Cross-Validation	AI Chat Assistant
Specialized analysis	70-95% reliability	HHH Framework	Follow-up questions

...

****How It Works:****

...

1. Ask Strategic Question
"Should we adopt AMD MI300X GPUs?"
2. Provide Context
Market data, constraints, competitive landscape
3. Get Analysis in 60s
Multi-agent report with confidence score

...

****Social Proof:****

- Customer logos (if available) or placeholder
- Testimonials: "Helped us make a \$70M allocation decision with confidence"
- Metrics: "247 analyses generated" "76% avg confidence" "4.6★ rating"

****CTA:****

- Primary: "Generate Your First Analysis" → `/new-analysis`
- Secondary: "View Sample Report" → `/demo`

2. New Analysis Page (`/new-analysis`)

Layout:
...

StratAI

[\[Profile\]](#) [\[Help\]](#)

New Strategic Analysis

Your Information

Email:

Company:

Role:

Strategic Question *

What decision are you facing?
(e.g., "Should we implement tiered
rate-limiting for GPU capacity?")

Context (optional but recommended)

Market | Strategic | Risk | Competitive |
Intel | Context | Factors | Landscape | Org

[Each opens textarea]

Market Signals

AI Chip Demand [=====●=====] 88%

Crypto Mining [==●=====] 12%

EV Components [===●=====] 20%

Supply Risk [=====●=====] 65%

Regulatory Risk [=====●=====] 55%

Consumer Electronics[=====●=====] 48%

Upload PDF (optional, max 10MB)

 Drag & drop or click to upload

[Cancel] [Generate Analysis →]

Form Validation:

- Email: Valid format required
- Company: 1-200 characters
- Strategic question: 10-500 characters (show counter)
- Market signals: 0-100 range with visual slider
- PDF: Max 10MB, .pdf only
- Show inline validation errors

****UX Details:****

- Autosave to localStorage (prevent data loss)
- Show character count for text fields
- Collapse/expand context sections
- Tooltips explaining each market signal
- Progress indicator: "Step 1 of 1" or tabs if multi-step
- Disable submit button until required fields valid

3. Analysis Progress Page (`/analysis/[id]/progress`)

Real-time Progress Display:

Analysing Your Strategic Question...

Current Status:

- ✓ Extracting context
- ✓ Parsing PDF document
- ✓ Analyzing market data
- ✓ Gathering competitive intelligence
- ⌚ Performing strategic analysis...
- ⌚ Assessing risks
- ⌚ Evaluating competitive position
- ⌚ Synthesizing insights
- ⌚ Calculating confidence score

A horizontal progress bar with a black dot indicating 75% completion. The bar is labeled 'Analysing Your Strategic Question...' at the start and '75%' at the end.

Estimated time remaining: ~20 seconds

💡 Did you know? Our 5 AI agents cross-validate their findings to ensure accuracy.

Technical Implementation:

- Poll `/api/analysis/[id]/status`` every 2 seconds
- Show smooth progress bar animation
- Display rotating tips while waiting
- Handle errors gracefully (show retry option)
- Redirect to report page when complete

4. Report View Page (`/report/[id]`)

Report Layout:

StratAI

Strategic Analysis Report

[Export ▼]

Company: Microsoft Azure

Question: Should we implement tiered rate-limiting?

Generated: Jan 18, 2025 10:30 AM

Confidence Score

72.45%

MODERATE-HIGH

Strong agreement among agents with minor conflicts on customer behavior predictions

Navigation

Executive Summary

Market Comparison

• Strategic Insights • Top 3 Recommendations • Risk Analysis • Agent Reliability									
Executive Summary									
[Formatted text with key findings highlighted]									
Market Comparison									
Signal	Rate	Market	Agents	Gap					
AI Chip	88%	High	Strong	Low					
Supply	65%	Risk	Concern	Med					
[Continue with other sections...]									
 Ask AI Assistant [Chat interface at bottom]									

- Features:**
- Sticky header with quick navigation
 - Confidence score prominently displayed with visual gauge
 - Expandable/collapsible sections
 - Copy-to-clipboard for sections
 - Export options: PDF, Google Docs link
 - Share via email/Slack
 - Print-friendly view

Confidence Score Visualization:

```

typescript
// Visual indicator based on score
score >= 75: Green gauge, "HIGH" label
score 60-75: Yellow gauge, "MODERATE" label
score 40-60: Orange gauge, "FAIR" label

```

score < 40: Red gauge, "LOW" label

```
// Include explanation
"This score reflects [strong/moderate/weak] agreement
among our 5 AI agents with [no/minor/major] conflicts."
'''
```

5. AI Chat Component

```
**Chat Interface.**
'''
```

Ask AI Assistant

You:

10:35 AM

Can you explain why the confidence score is 72%?

StratAI:

10:35 AM

The 72.45% confidence score reflects moderate agreement among our 5 AI agents with some conflicting views on customer retention...

[Full response with citations to report sections]

Suggested questions:

• What are the biggest risks identified?

• How reliable is the Market Data agent?

• What would increase the confidence score?

Type your question...

Send →

```
**Chat Features.**
- Real-time streaming responses (if possible)
- Message history persists per report
```


- Suggested follow-up questions
- Links to relevant report sections
- Copy message to clipboard
- Timestamp for each message
- Loading indicator while AI responds
- Error handling with retry option

6. Feedback Modal

Feedback Collection:

...

Help Us Improve

[X]

How would you rate this analysis?

★

★

★

★

★

(Click to rate)

Was this analysis useful?

Yes, very useful

Somewhat useful

Not useful

Would you recommend StratAI to a colleague?

Definitely

Probably

Not sure

Probably not

Additional feedback (optional):

Tell us more about your experience...

[Skip]

[Submit Feedback]

...

Trigger:

- Auto-show after user spends 2+ minutes on report
- Or manually via "Provide Feedback" button
- Only show once per report

7. Admin Dashboard (`/admin`) - Optional MVP

Analytics Overview:

...

StratAI Admin

Last 30 days

247

76.84%

4.6★

89%

Reports

Avg

Rating

Would

Conf.

Recom.

Reports Generated (Last 7 Days)

Mon Tue Wed Thu Fri Sat Sun

Top Companies

1. Microsoft Azure

45 reports

2. Google Cloud

38 reports

3. Oracle Cloud

22 reports

Recent Reports

ID

Company

Conf.

Date

RPT-1234

Azure

72.5%

10:30 AM

[View]

RPT-1233

Oracle

85.2%

9:15 AM

[View]

RPT-1232

GCP

68.9%

Yesterday

[View]

API Integration

Backend Endpoints

Create Analysis:

typescript

POST /api/analysis

Body: {

```
  user_email: string;
  company_name: string;
  job_function: string;
  strategic_question: string;
  market_intelligence?: string;
  strategic_context?: string;
  risk_factors?: string;
  competitive_landscape?: string;
  organizational_insights?: string;
  user_ai_chip_rate: number;    // 0-100
  user_crypto_rate: number;    // 0-100
  user_ev_rate: number;        // 0-100
  user_supply_risk: number;    // 0-100
  user_regulatory: number;     // 0-100
  user_consumer_rate: number;  // 0-100
  pdf_file?: string;          // base64
```

}

Response: {

```
  success: true,
  analysis_id: string,
  estimated_time: number // seconds
```

}

Check Analysis Status:

typescript

GET /api/analysis/:id/status

Response: {

```
  status: 'processing' | 'completed' | 'failed',
```

```
    progress: number,    // 0-100
    current_step: string,
    estimated_remaining: number
}
```

Get Report:

typescript

GET /api/analysis/:id/report

```
Response: {
  report_id: string,
  confidence_score: string, // "72.45"
  full_report: string,    // Markdown
  executive_summary: string,
  recommendations: string,
  // ... other sections
  google_doc_url: string,
  created_at: string
}
```

Chat with AI:

typescript

POST /api/chat

```
Body: {
  report_id: string,
  message: string,
  conversation_history?: Array<{role: string, content: string}>
}
```

```
Response: {
  response: string,
  timestamp: string
}
```

Submit Feedback:

typescript

POST /api/feedback

```
Body: {
  report_id: string,
  rating: number,    // 1-5
}
```

```
    useful: boolean,
    would_recommend: boolean,
    feedback_text?: string
  }
```

```
Response: {
  success: true,
  feedback_id: string
}
```

Get Analytics (Admin):

typescript

GET /api/admin/analytics

```
Response: {
  total_reports: number,
  avg_confidence: string,
  avg_rating: string,
  reports_by_company: Record<string, number>,
  recent_reports: Array<{
    report_id: string,
    company: string,
    confidence: string,
    timestamp: string
  }>
}
```

Technical Requirements

Framework & Libraries

- **React 18+** with TypeScript
- **Next.js 14** (App Router) or **Vite**
- **Tailwind CSS** for styling
- **shadcn/ui** for component library
- **Recharts** or **Chart.js** for visualizations
- **React Query** for data fetching
- **Zustand** or **Context API** for state management
- **React Hook Form** + **Zod** for form validation
- **Lucide React** for icons
- **React Markdown** for report rendering

Performance

- Lazy load heavy components
- Optimize images with Next/Image
- Code splitting for routes
- Implement skeleton loaders
- Cache API responses
- Debounce search/filter inputs

Accessibility

- WCAG 2.1 AA compliance
- Semantic HTML
- ARIA labels where needed
- Keyboard navigation
- Focus management
- Color contrast ratios met
- Screen reader tested

Responsive Design

- Mobile-first approach
- Breakpoints: sm (640px), md (768px), lg (1024px), xl (1280px)
- Touch-friendly tap targets (min 44x44px)
- Optimized layouts for tablet
- Hamburger menu on mobile

SEO & Meta

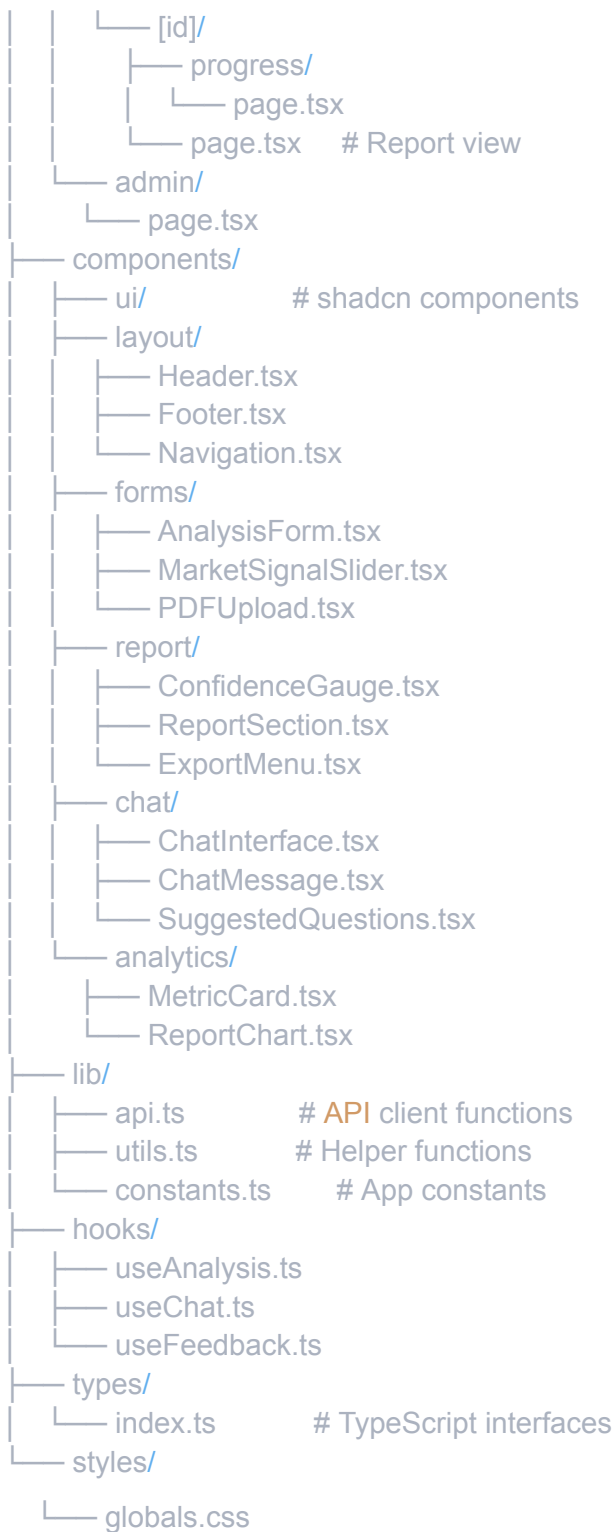
- Dynamic meta tags per page
- OpenGraph tags for sharing
- Structured data (JSON-LD)
- Sitemap generation
- robots.txt configuration

Component Architecture

Suggested File Structure

...

```
src/
├── app/                # Next.js App Router
│   ├── page.tsx        # Landing page
│   ├── new-analysis/
│   │   └── page.tsx
│   └── analysis/
```



Key Components

ConfidenceGauge.tsx

typescript

```
interface ConfidenceGaugeProps {  
  score: number; // 0-100  
  size?: 'sm' | 'md' | 'lg';  
}
```

// Visual circular gauge with color coding

// Green (>=75), Yellow (60-74), Orange (40-59), Red (<40)

MarketSignalSlider.tsx

typescript

```
interface MarketSignalSliderProps {  
  label: string;  
  value: number;  
  onChange: (value: number) => void;  
  description?: string;  
}
```

// Range slider 0-100 with tooltip on hover

ChatInterface.tsx

typescript

```
interface ChatInterfaceProps {  
  reportId: string;  
}
```

// Real-time chat with AI

// Message history, suggested questions, streaming responses

User Experience Details

Loading States

- Skeleton screens for reports
- Spinning indicators for API calls
- Progress bars for analysis
- Shimmer effects for placeholders

Error Handling

typescript

// Toast notifications for errors

```
toast.error("Analysis failed. Please try again.");
```

// Inline validation errors

```
<Input error="Email is required" />
```

// Full-page error states with retry

```
<ErrorBoundary>
```

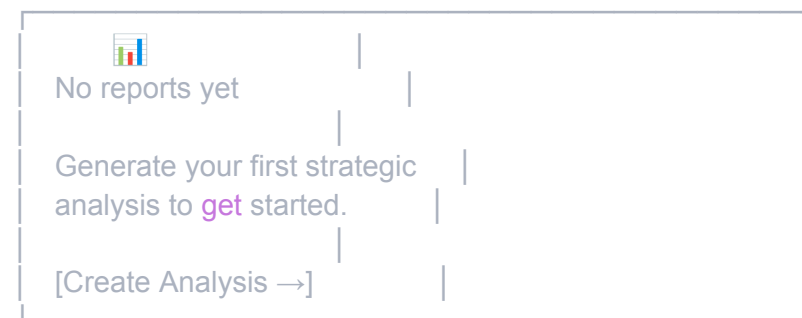
```
  <ReportView />
```

```
</ErrorBoundary>
```

```
...
```

Empty States

```
...
```



```
...
```

Success States

```
...
```



Sample Data for Development

Mock Analysis Response:

```
json
{
  "report_id": "RPT-1737158400123",
  "confidence_score": "72.45",
  "strategic_question": "Should we implement tiered rate-limiting?",
  "company_name": "Microsoft Azure",
  "executive_summary": "Based on multi-agent analysis, implementing tiered rate-limiting is recommended with moderate-high confidence (72.45%). This approach balances customer satisfaction with capacity management...",
  "recommendations": "1. Implement three-tier system\n2. Grandfather top 10 customers\n3. Monitor satisfaction metrics closely",
  "created_at": "2025-01-18T10:30:45.123Z",
  "google_doc_url": "https://docs.google.com/document/d/abc123"
}
```

Mock Chat Response:

```
json
{
  "response": "The 72.45% confidence score reflects moderate agreement among our 5 AI agents. The Market Data and Strategic Analysis agents showed strong alignment (85% and 82% HHH scores), while the Risk Management agent had some uncertainty about customer behavior predictions (68% HHH). This creates a base score of 74.45% with a -2% adjustment for minor conflicts, resulting in the final 72.45%.",
  "timestamp": "2025-01-18T10:35:12.456Z"
}
```

Deployment Checklist

- Environment variables configured
- API endpoints tested
- Error boundaries implemented
- Loading states for all async operations
- Form validation working
- Mobile responsive design verified
- Accessibility audit passed
- SEO meta tags configured
- Analytics tracking setup (optional)
- Performance optimized (Lighthouse >90)

Nice-to-Have Features (Post-MVP)

1. **Dark mode** toggle
 2. **Report comparison** - side-by-side view of multiple reports
 3. **Saved analyses** - bookmark and organize reports
 4. **Team collaboration** - share reports with team members
 5. **Email notifications** - when analysis completes
 6. **Report templates** - pre-fill common scenarios
 7. **Advanced filtering** in admin dashboard
 8. **Export to PowerPoint** slides
 9. **Scheduled analyses** - recurring strategic reviews
 10. **Mobile app** (React Native)
-

Brand Voice & Copywriting

Tone: Professional, confident, data-driven, empowering

Avoid: Overly technical jargon, hype, vague promises

Example Copy:



Good:

"Make strategic decisions with confidence. Our multi-agent AI analyzes your question from 5 perspectives and shows you exactly how confident you should be in the recommendation."



Bad:

"Revolutionary AI technology disrupts decision-making with next-gen algorithms!"

Button Labels:

- "Generate Analysis" (not "Submit")
- "View Report" (not "See Results")
- "Ask AI Assistant" (not "Chat")
- "Export Report" (not "Download")

Microcopy:

- Form helper text: "Share your market constraints, customer dynamics, and organizational context for richer analysis."
- Empty state: "No reports yet. Generate your first strategic analysis to get started."

- Error: "Analysis failed. Our team has been notified. Please try again or contact support@stratai.com."
-

Final Notes

This is a **B2B executive tool** - prioritize clarity, trust, and professionalism over flashy animations. The confidence score is the hero feature - make it prominent and easy to understand.

The target user is a **busy director or VP** who needs answers quickly but wants to see the reasoning. Design for scanning, not deep reading (though the full report should be available).

Test with real strategic questions from the semiconductor/hardware industry to ensure the UI handles complex, domain-specific language gracefully.